Problem

Edit Distance

S1 = reddit i

S2 = edit j

replace $(i-1, j-1)$

$(i-1, j)$ remove

insert $(i, j-1)$

reedit
edit

3

reddit
edit

2

rededit
edit

4

insert example

0 1 2 3 4 5
r e d d i t
i

0 1 2 3
e d i t
j

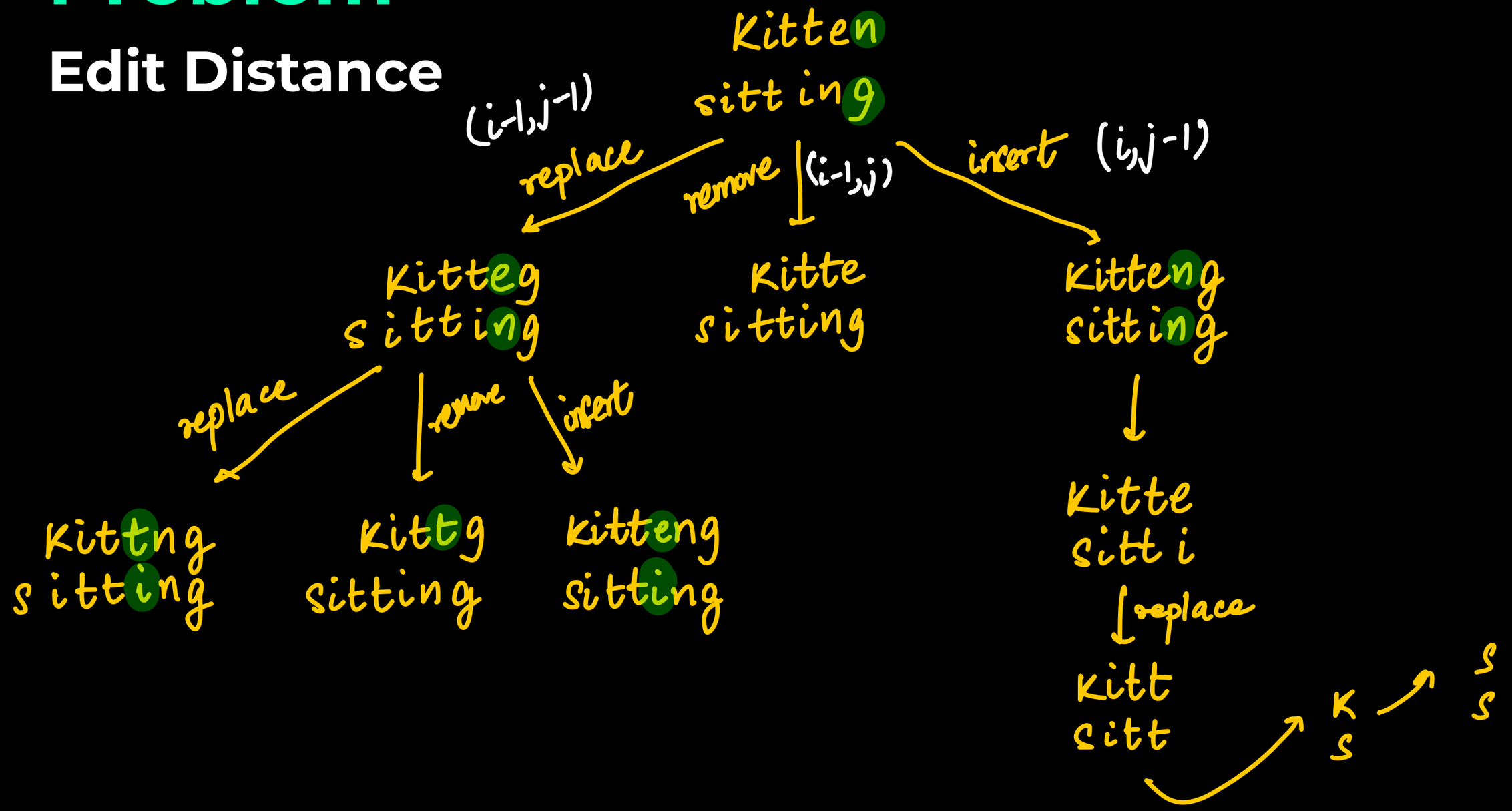
⇒

0 1 2 3 4 5
red e dit
i

0 1 2 3
edit
j

Problem

Edit Distance



Problem

s1 = anuj
s2 = anjum

anu , anj 3

Longest Common Subsequence (Tabulation)

		a	n	j	u	m
		0	1	2	3	4	5
	0	0	0	0	0	0	0
a	1	0	1	1	1	1	1
n	2	0	1	2	2	2	2
u	3	0	1	2	2	3	3
j	4	0	1	2	3	3	3

Problem

S1 = anuj
S2 = anjum

anu , anj 3

Print Longest Common Subsequence

		a	n	j	u	m
		0	1	2	3	4	5
	0	0	0	0	0	0	0
a	1	0	1	1	1	1	1
n	2	0	1	2	2	2	2
u	3	0	1	2	2	3	3
j	4	0	1	2	3	3	3

Same pe left

ans = jna
↓reverse
anj

Problem

Print Longest Common Subsequence (print any one)

		a	n	j	u	m
		0	1	2	3	4	5
	0	0	0	0	0	0	0
a	1	0	1	1	1	1	1
n	2	0	1	2	2	2	2
u	3	0	1	2	2	3	3
j	4	0	1	2	3	3	3

Same me upar

ans = una

↓

anu

Problem

Shortest Common Supersequence

length

s1.len + s2.len − lcs(s1, s2)

s1 = raghav

s2 = raman

lcs = r a a

scs = r a g h m a v n

scs = r a m g h a v n

r a m g h a n v

r a g h m a v n

r a g h m a n v

r a g m h a n v

Problem

Shortest Common Supersequence

Kitten ^{i}

sitting ^{j}

LCS = ittn ^{k}

ans = ksitteing